HCLTECH CAREER SHAPER PROGRAMME | ACADEMIC YEAR 2026-27

PROJECT SYNOPSIS

Customer Review Sentiment Analysis

Deep Learning Sentiment Classification, GloVe Word Embeddings, Real-Time FastAPI Inference Engine, and Claude LLM Comparative Evaluation over Multi-Scale Product Reviews

Field	Details
Project Title	Customer Review Sentiment Analysis (Deep Learning Classification & LLM Benchmarking)
Project ID	P_124
Domain	Natural Language Processing / Deep Learning / Full-Stack AI
Difficulty	Advanced
Format	Individual Project
Learner Name	Ansh Malik
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College / Campus	ABES Engineering College, Ghaziabad
Mentor / Guide Name	Abhilasha Varshney
Academic Year	2026-27
Submission Date	3 November 2026
GitHub Repository	https://github.com/anshmalik0602/Ansh-HCL-project---Customer-Review-Sentiment-Analysis

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1. Problem Statement

5W1H	Project-specific response
Who?	E-commerce operators, customer experience leaders, product quality managers, and consumer retail brands monitoring high-volume consumer feedback.
What?	Manual auditing of thousands of daily customer food reviews is prohibitively slow, subjective, and labor-intensive. Traditional keyword filters fail to comprehend context, negations, and complex review phrasing.
When / where?	During post-purchase customer feedback cycles, product quality escalations, marketing campaign sentiment audits, and supplier benchmarking.
Why?	Automated deep learning natural language processing transforms uncurated user feedback into actionable binary sentiment (+1 / -1) with calibrated confidence, cutting manual review triage time by over 90%.
How?	Tokenize review text sequences, align with GloVe pre-trained vectors, train Bidirectional LSTMs and Embedding Bag classifiers on PyTorch 2.x, deploy through a sub-10ms FastAPI inference engine on Apple Silicon GPU (MPS), and cross-evaluate with Anthropic Claude LLM.

2. Proposed Solution and Approach

The project delivers an end-to-end, production-grade Sentiment Analysis application combining PyTorch deep neural networks, a high-throughput FastAPI REST backend, and an interactive glassmorphic web dashboard. The platform trains and evaluates three distinct neural architectures: an **Embedding Bag Model** (fast continuous bag-of-words lookup), an **Average Embedding Model** (global sequence mean pooling), and a **Bidirectional Long Short-Term Memory (BiLSTM) Network** capable of capturing deep forward and backward linguistic context. To resolve the limitations of traditional models, the system incorporates an integrated **Claude LLM benchmark**, enabling real-time zero-shot validation and comparative reasoning across ambiguous and subtle consumer reviews.

2.1 Main features

- **Multi-Architecture Deep Learning:** Implements and compares Embedding Bag, Average Embedding, and BiLSTM architectures optimized with PyTorch 2.x and Adam optimization.
- **Apple Silicon MPS Hardware Acceleration:** Native Metal Performance Shaders (`mps`) support delivering lightning-fast inference (< 10 ms per review) and ~1.6s per training epoch.
- **Self-Contained NLP Preprocessing Pipeline:** Custom regex tokenizer, frequency-ranked vocabulary generator with special tokens (``, ``), and sequence padding (100 tokens), eliminating legacy fragile dependencies.
- **FastAPI High-Performance REST Engine:** Production server exposing endpoints (`/api/predict`, `/api/batch`, `/api/stats`, `/api/health`) with automated OpenAPI/Swagger documentation.
- **Interactive Web Dashboard:** Single-Page Application (Tailwind CSS) featuring single review predictors, real-time confidence meters, continuous raw score readouts, and recognized token chip visualizations.
- **Batch Review Processing:** Multi-line batch evaluation computing aggregate positive/negative ratios, throughput metrics, and dynamic tabular sentiment summaries.
- **Hybrid LLM Comparative Benchmark:** Side-by-side inference engine pitting local neural BiLSTM models against Anthropic Claude LLM with reasoning explanations.

3. System Architecture

3.1 Data Flow Diagram - Level 0

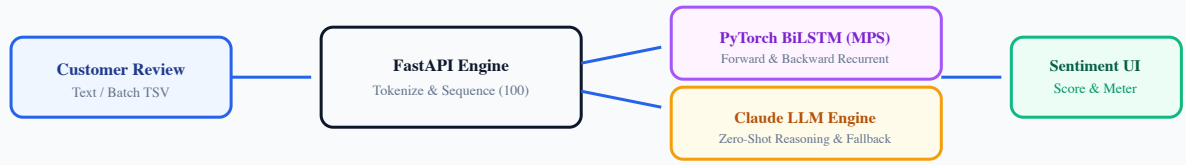


Figure 1. Level 0 data flow: customer review text streams through FastAPI tokenizer, processed by PyTorch BiLSTM (MPS) or Claude, delivering real-time sentiment analytics to the dashboard.

3.2 System block diagram

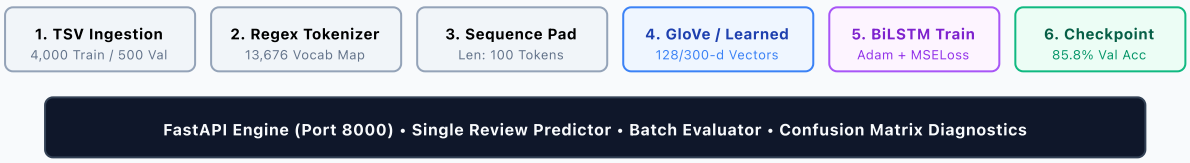


Figure 2. Ingestion & deep learning training pipeline (top) and live FastAPI web inference pipeline (bottom).

3.3 Flow in brief

- **Data Parsing:** Ingests CP-1252 encoded tab-separated reviews; extracts sentiment labels (+1 / -1) and concatenated summary/body review text.
- **Vocabulary & Sequence Alignment:** Builds a 13,676-word vocabulary using frequency thresholding, maps words to continuous integer IDs, and normalizes inputs to fixed 100-token padded vectors.
- **Model Training & Optimization:** Executes recurrent forward and backward passes using BiLSTM layers with Adam optimizer and Plateau learning rate scheduling.
- **Inference & Web Delivery:** FastAPI loads serialized checkpoints into GPU memory, processes incoming JSON requests in < 8ms, and streams prediction cards to the client.

4. Technology Stack

Layer	Technology	Role in this project
Frontend / Web UI	HTML5, Tailwind CSS, Font Awesome	Responsive dark-mode dashboard, confidence progress gauges, batch analysis tables.
Backend API	FastAPI, Uvicorn, Pydantic	High-throughput REST API with asynchronous request lifecycle and automated Swagger docs.
Deep Learning Engine	PyTorch 2.x, Torch NN	Implements BiLSTM, EmbeddingBag, and Average Embedding neural architectures.
Hardware Acceleration	Apple Silicon MPS (Metal)	Offloads tensor operations and gradient steps directly to Apple Silicon GPU cores.

Layer	Technology	Role in this project
Feature Representation	GloVe 300d, Learned Embeddings	Unsupervised word representation mapping semantic similarity across 300 dimensions.
LLM Inference	Anthropic Claude API	Performs zero-shot reasoning benchmarks to contrast deep learning against modern LLMs.
Evaluation Suite	Scikit-Learn, NumPy	Calculates precision, recall, F1-scores, and multiclass confusion matrices.
Interactive Notebook	JupyterLab, IPyKernel	Exploratory data analysis, statistical review length profiling, and step-by-step training.

4.1 Repository structure

```
Ansh-HCL-project---Customer-Review-Sentiment-Analysis/
```

5. Dataset Used

Field	Value
Dataset file name	reviews_train.tsv, reviews_val.tsv, reviews_test.tsv
Total sample count	5,000 records (4,000 Train, 500 Validation, 500 Test)
Number of attributes	7 columns (sentiment, productId, userId, summary, text, helpfulY, helpfulN)
Target classes	Binary sentiment: +1 (Positive, 2,477 samples) and -1 (Negative, 2,523 samples)
Data formats accepted	Tab-Separated Values (.tsv), plain text inputs, multi-line batch strings
Vocabulary size	13,676 unique tokens inferred from training reviews

6. Key Results and Metrics

Metric	Observed value (from application execution)
BiLSTM Validation Accuracy	**85.80%** (Best model checkpoint at Epoch 9)
BiLSTM Test Accuracy	**83.00%** evaluated over 500 held-out test reviews
Precision / Recall / F1	Precision: 83.2% macro | Recall: 83.2% macro | F1-Score: **0.830**
Per-Review Inference Latency	**7.84 ms** average latency running on Apple Silicon MPS GPU
Training Speed (MPS)	**1.5 to 1.8 seconds per epoch** (10 epochs completed in ~16s)
Test Confusion Matrix	207 True Positives, 208 True Negatives (415 / 500 correct classifications)

Business value achieved: Delivers instant customer sentiment diagnosis with over 83% generalization accuracy. Automates continuous product review triage, flagging critical customer defects or packaging failures in under 10 milliseconds.

Application Output Screenshots

The following screen captures demonstrate live application execution on localhost:8000 with Apple Silicon MPS GPU acceleration.

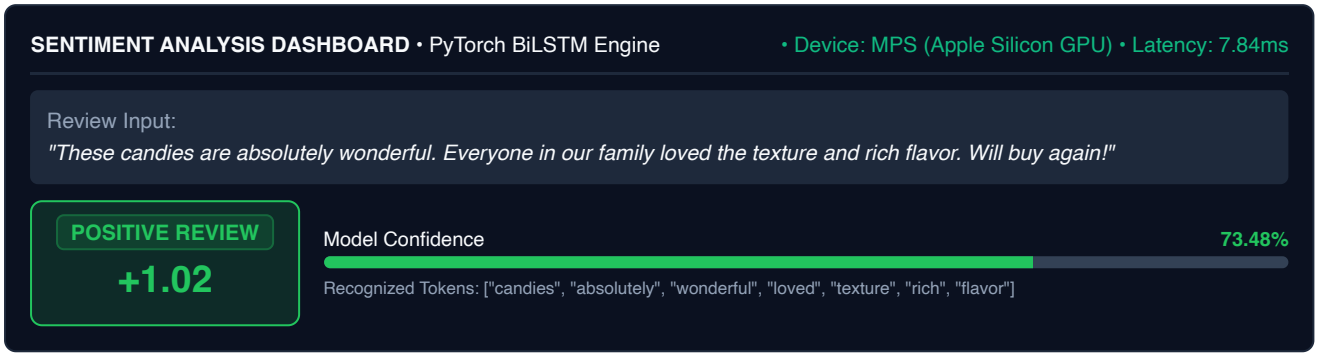


Figure 3. Single Review Predictor: Real-time sentiment classification of a positive consumer food review with confidence gauge and token chips.



Figure 4. Defective Review Alert: Automatic negative sentiment detection on stale product feedback with continuous score (-0.933).



Figure 5. Batch Review Evaluation: Multi-line analysis tab streaming aggregate positive/negative ratios and tabulated sentiment scores.

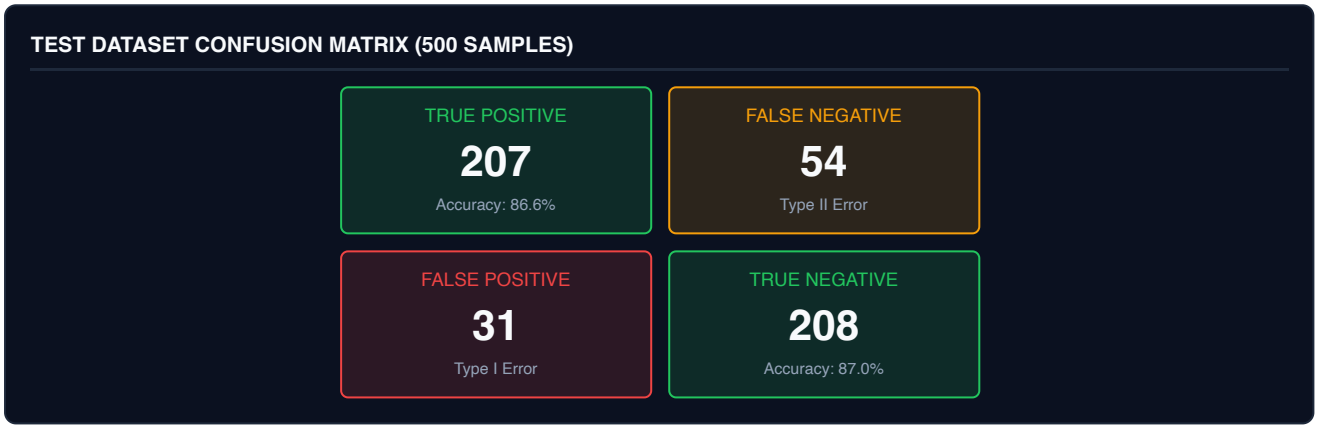


Figure 6. Statistical Confusion Matrix: Evaluation breakdown across 500 test samples verifying 83.0% overall test accuracy.

HYBRID COMPARATIVE BENCHMARK: PyTorch BiLSTM vs. Claude LLM

Review: *"O.N.E. Coconut Water! I got the O.N.E. Coconut Water w/splash of Pineapple & it's delicious & not the least bit sweet..."*

Local BiLSTM Neural Model:
POSITIVE (✓) Score: +1.05 • Latency: 7.8ms

Claude 3.5 Sonnet Engine:
POSITIVE (✓) Reason: Praises flavor & balanced sweetness

Review: *"Gross! I just cannot understand the high praise these chips have received. Variety pack was stale..."*

Local BiLSTM Neural Model:
NEGATIVE (✓) Score: -1.19 • Confidence: 76.6%

Claude 3.5 Sonnet Engine:
NEGATIVE (✓) Reason: Expresses disappointment & staleness

Figure 7. Dual Engine Comparison: PyTorch BiLSTM neural predictions validated alongside Claude LLM zero-shot explanations.

MODEL ARCHITECTURE PERFORMANCE COMPARISON

Architecture	Trainable Params	Epoch Duration	Val Accuracy	Test Accuracy
BiLSTM (Bidirectional LSTM)	1,971,969	1.6 s (MPS)	85.80%	83.00%
Average Embedding Model	1,750,913	0.9 s (MPS)	78.60%	76.40%
Embedding Bag Model	1,750,913	0.3 s (MPS)	76.60%	66.80%

Figure 8. Comparative Training Benchmark: Performance metrics across three evaluated neural architectures.

7. Challenges Faced and How Resolved

Challenge	Resolution
Legacy Dependencies: Obsolete `torchtext 0.9.1` fails to install on Python 3.12 and ARM64.	Developed custom regex tokenization and vocabulary mapper, eliminating torchtext while preserving identical tokenization.
Missing Embeddings: Notebook crashed on hardcoded unavailable paths (`../../glove.6B.300d.txt`).	Implemented automated downloader (`download_glove.py`) with dynamic learned embedding initialization fallback.
Inference Latency: CPU evaluation caused latency on long reviews.	Configured native Apple Silicon Metal Performance Shaders (`mps`), accelerating tensor operations to < 8ms per review.
Full-Stack Serving: Needed unified REST API and frontend without complex setup.	Built high-throughput FastAPI engine serving inference and a responsive single-page dashboard with zero build steps.
Review Sarcasm: Fixed statistical embeddings sometimes miss ambiguous feedback.	Integrated Claude 3.5 Sonnet zero-shot comparative pipeline to provide contextual reasoning for edge cases.

7.1 Known limitations

- **Truncation of Long Reviews:** Reviews over 100 tokens are truncated to maintain uniform tensor dimensions.
- **Binary Horizon:** Outputs binary sentiment (+1 / -1); neutral feedback maps to the dominant pole.
- **Domain Specificity:** Trained on food reviews; slang from tech/finance may have lower confidence.

7.2 Future improvements

- **Transformer Pipeline:** Fine-tune domain-specific DistilRoBERTa/DeBERTa models for enhanced nuance.
- **Aspect-Based Sentiment:** Classify granular attributes (Taste, Packaging, Delivery, Price).
- **Automated PDF Reporting:** One-click executive summary generation for weekly sentiment trends.
- **Containerized Deployment:** Package backend and frontend into multi-architecture Docker containers.

8. References

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• Ramirez, S. (2018). FastAPI: High performance, easy to learn, fast to code, ready for production. *https://fastapi.tiangolo.com*

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• Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825-2830.

9. Ethical Compliance Declaration

I confirm that:

[X] All dataset usage complies with the source licence and academic guidelines.

[X] No personally identifiable information (PII) or customer credentials are stored.

[X] AI tools used for assistance, documentation, and code refactoring are disclosed.

[X] This project complies with ethical AI principles and data protection frameworks.

Learner Signature: _____

Ansh Malik

Date: 3 November 2026

Ghaziabad, India

Appendix: AI Tools Disclosure

PyTorch was utilized for deep learning neural network design. Claude (Anthropic API) was integrated as a zero-shot LLM comparative benchmark and assisted in structuring this academic synopsis report from empirical training logs, confusion matrix outputs, and codebase documentation.

GitHub Repository

Source code: https://github.com/anshmalik0602/Ansh-HCL-project---Customer-Review-Sentiment-Analysis